

PAUL MITCHELL

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TECHNOLOGY CONSULTING & SOLUTIONS ENGINEERING ANALYST

PROFESSIONAL SUMMARY

William & Mary Computer Science M.S. candidate (3.95 GPA, AI/ML & Systems concentration) and B.S. graduate with hands-on experience in Linux systems administration, cloud infrastructure, enterprise telemetry, and pre-inference AI defense architectures. Proven ability to evaluate complex technical trade-offs, orchestrate configuration automation across 100+ production nodes, and translate mission-critical technical architectures into actionable operational decisions for executive and client stakeholders.

EDUCATION

College of William & Mary — M.S. in Computer Science (AI/ML & Systems) Expected Dec 2026 · GPA: 3.95
College of William & Mary — B.S. in Computer Science Graduated Dec 2024
Relevant Coursework: Machine Learning Systems, Advanced Software Engineering, Deep Learning, Distributed Systems, Data Mining

TECHNICAL SKILLS & INFRASTRUCTURE

Cloud & Infrastructure: AWS (S3, EC2, Bedrock), Docker, Kubernetes, Linux (Ubuntu/RHEL), SSH, DNS, Netplan, Puppet, system hardening
Observability & Monitoring: Prometheus, Grafana, OpenTelemetry, alert rule design, incident triage, log auditing, capacity planning
Languages & Scripting: Python, FastAPI, Java, Bash, PowerShell, C/C++, JavaScript, SQL (PostgreSQL), YAML, Git/GitLab CI
Systems & Security: Pre-inference LLM defense, zero-width steganography, STIX 2.1, ArcSight CEF, anomaly detection, threshold calibration

TECHNICAL & PROFESSIONAL EXPERIENCE

IT Administrator | William & Mary Technology Support Center | Williamsburg, VA Feb 2025 – Present

- Administer, troubleshoot, and maintain over 100 Linux instructional workstations and compute servers, resolving network routing, DNS, storage, account, and application incidents.
- Orchestrated centralized configuration automation and host provisioning via Puppet, eliminating configuration drift and standardizing security baseline policies across multi-room labs.
- Built and deployed Prometheus and Grafana telemetry dashboards monitoring GPU allocation, compute metrics, and storage pools, cutting downtime response times by over 40%.
- Conducted architectural evaluation of multi-user Docker containerization, authoring formal technical trade-off documentation on isolation boundaries and operational maintainability.
- Standardized network configurations and maintained structured inventory records in Excel, standardizing hardware lifecycles and static IP assignments.

Data Analytics & AI Engineering Extern | Extern (Amazon-Sponsored Project) | Remote Jun 2025 – Sep 2025

- Engineered automated, reproducible Python and AWS S3 data pipelines to ingest, clean, and validate ~20,000 text records, reducing recurring data prep cycles from hours to minutes.
- Fine-tuned DistilBERT classification models ($F1 = 0.89$) and applied BERTopic clustering to isolate 5 root causes of operational turnover from unstructured employee feedback.
- Trained predictive LightGBM decision models and deployed an interactive Streamlit telemetry dashboard delivering real-time decision-support metrics for program mentors.
- Implemented cloud dataset versioning using DVC on AWS S3 to ensure reproducibility and audit compliance across multi-team evaluations.

KEY TECHNICAL & SYSTEMS PROJECTS

ShadowPrompt: Pre-Inference Defense Proxy for Enterprise LLMs | Python, FastAPI, Three.js 2026 · pjmittchell7.github.io/shadowprompt

- Architected an in-memory pre-inference proxy inspecting raw Unicode streams at 0.038ms P99 latency, blocking steganography, directional overrides, and template breakouts before tokenization.
- Integrated deceptive honeypot sandboxes seeded with canary credentials to waste adversary compute without triggering evasion routines; exported compliant STIX 2.1 and ArcSight CEF syslog feeds.

PromptShield: Tokenization-Aware LLM Security Middleware | Python, FastAPI, scikit-learn 2026

- Built asynchronous FastAPI middleware intercepting adversarial prompt injections with 0.27–1.11ms latency, avoiding expensive secondary LLM calls while holding false positives strictly under 0.1%.
- Engineered calibrated threshold scoring against production text streams, validating model performance on complex technical dialogue.

Healthcare Network Traffic Anomaly Detection Framework | Python, scikit-learn 2025

- Developed an automated network packet inspection pipeline featuring feature extraction and threshold calibration to support SOC operational alerting.
- Trained ensemble classifiers isolating anomalous flow signatures with 99.1% detection precision under high-throughput conditions.